UNIT - III
Industrial Robotics, Actuators & Sensors

Automation and Robotics (23A03606) - handwritten study notes

1. Introduction to Industrial Robotics

1.0 Syllabus topics covered

- Industrial robotics, configurations, functional line diagram, DOF, arms, joints, grippers, gripper design
- Actuators: pneumatic, hydraulic, electric, stepper; position, velocity, tactile and proximity sensors

1.1 What is a robot?

- The word **robot** comes from the Czech word **robota** meaning forced labour. It was used by Karel Capek in his play R.U.R. (1920)
- The word **robotics** was coined by Isaac Asimov (1940s) who also gave the three laws of robotics
- **RIA definition (Robot Institute of America):** A robot is a reprogrammable, multifunctional manipulator designed to move materials, parts, tools or specialised devices through variable programmed motions for the performance of a variety of tasks.
- **ISO 8373 definition:** An automatically controlled, reprogrammable, multipurpose manipulator, programmable in three or more axes, which may be fixed in place or mobile, for use in industrial automation applications.

Three key words in every definition: **reprogrammable**, **multifunctional** and **manipulator**. A fixed-sequence machine with no reprogramming is NOT a robot.

1.4 Basic terms

Term	Meaning
Manipulator	Arm and wrist assembly made of links and joints that positions the end effector
End effector	Tool or gripper fixed at the wrist to do the work
Payload	Maximum weight the robot can carry at full speed and still meet its specification
Work envelope	Set of all points in space that the wrist/end effector can reach
Reach	Maximum distance from the base axis to the wrist
Repeatability	How closely the robot returns to a previously taught point (smaller value is better)
Accuracy	How close the robot gets to a commanded point in space
Resolution	Smallest incremental move the control system can command
Speed	Maximum velocity of the tool centre point (TCP)

2. Classification of Robots

3. Classification of Robot Configurations

The first three joints (the **arm** **and** **body**) decide the shape of the work envelope. The last three joints (the **wrist**) decide the orientation of the tool. Joints are either linear (L or P - prismatic) or rotary (R - revolute).

3.1 Cartesian (rectangular / gantry) - LLL or PPP

- Three linear axes X, Y, Z at right angles to each other
- Work envelope is a **rectangular** **box**
- Advantages: simple control, very high accuracy and rigidity, easy to program, large payload in gantry type
- Limitations: needs a lot of floor space, small reach inside the machine, axes are hard to seal
- Uses: pick and place, CNC loading, assembly, 3D printing

3.3 Spherical (polar) - RRL (TRL)

- One linear telescoping arm that can pivot about a horizontal axis and rotate about a vertical base axis
- Work envelope is a **part of a sphere**
- Advantages: long reach, can reach down to the floor level
- Limitations: low accuracy at long extension, complex control
- Uses: die casting, forging (Unimate and Versatran were polar robots)

3.4 Articulated (jointed arm / anthropomorphic) - RRR

- Three rotary joints: waist, shoulder and elbow, like a human arm
- Work envelope is an **irregular sphere-like volume**
- Advantages: maximum flexibility, compact, large work volume, can reach over and around obstacles
- Limitations: complex control and kinematics, low stiffness at full stretch, accuracy changes with position
- Uses: arc welding, spray painting, assembly - the most common industrial robot

3.7 Comparison of configurations

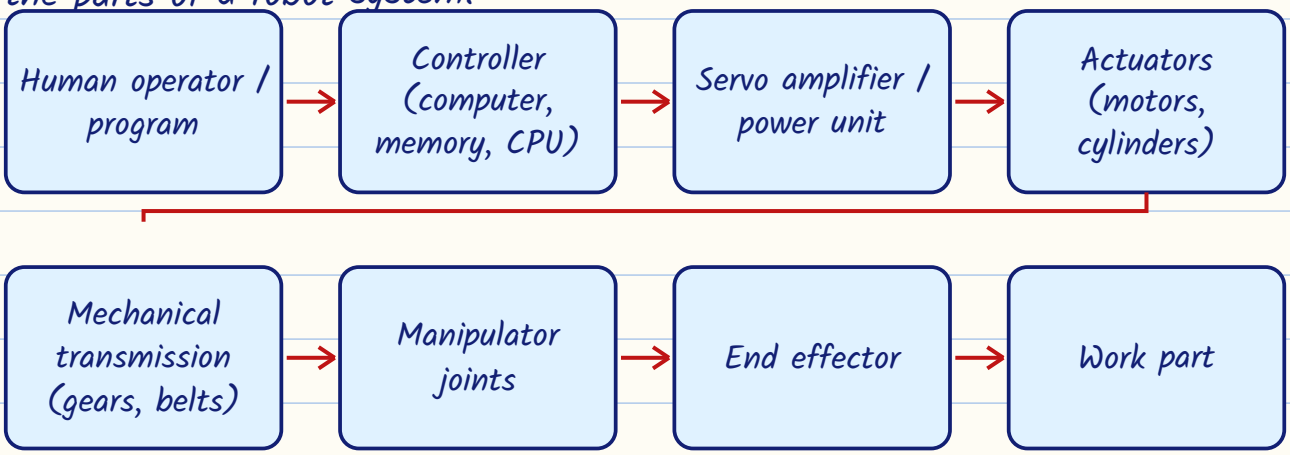
Type	Joints	Work envelope	Main use
Cartesian	PPP	Rectangular box	Pick and place, gantry
Cylindrical	RPP	Cylinder	Machine loading
Spherical	RRP	Sphere segment	Forging, die casting
Articulated	RRR	Irregular sphere	Welding, painting
SCARA	RRP	Cylinder-like (horizontal)	Assembly

Cartesian X Y Z → Cylindrical R Z r → Spherical R R r → Articulated R R R

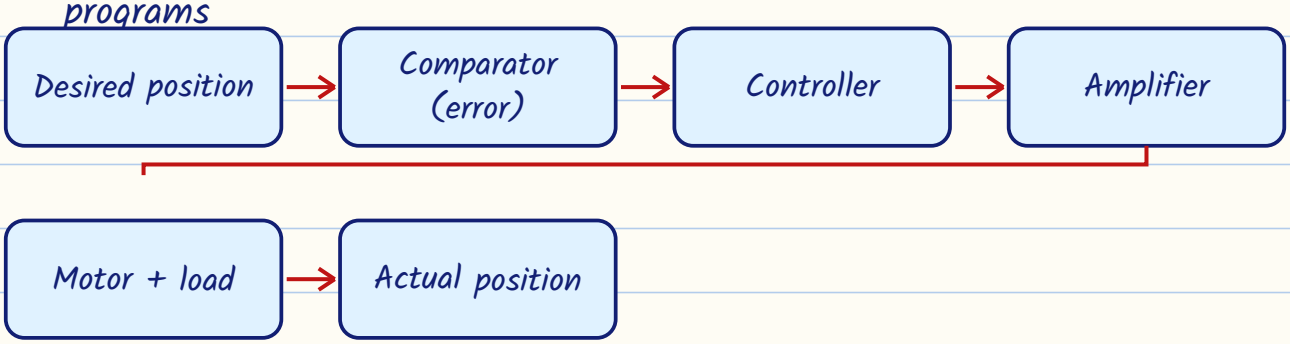
Memory trick: the number of rotary joints rises from 0 to 3 as we go Cartesian (0) ⟶ Cylindrical (1) ⟶ Spherical (2) ⟶ Articulated (3).

4. Functional Line Diagram of a Robot

The functional line diagram shows how information and energy flow between the parts of a robot system.



- Feedback path: **sensors** on every joint measure position and speed and return the signal to the controller. This closed loop gives accuracy
- Position sensors (encoder, resolver, potentiometer) give joint angle; tachogenerator gives velocity
- External sensors (vision, force, touch, proximity) tell the controller about the work environment
- The teach pendant is used by the operator to teach points and edit programs



- The comparator subtracts the **feedback** signal from the **command** signal. The error drives the motor until the error is zero

> A robot with feedback is a closed loop (servo) system. A robot without feedback (stepper driven, pneumatic stops) is an open loop system.

5. Degrees of Freedom (DOF)

Degree of freedom is an independent motion (translation or rotation) that a mechanism can make. Each joint of a robot gives it one DOF (one axis). In space a free rigid body has **6 DOF**: three translations (x, y, z) and three rotations (roll, pitch, yaw).

- To place a tool at any position and orientation in space a robot needs at least **6 DOF**
- First 3 DOF (arm and body) position the wrist. Last 3 DOF (wrist) orient the tool
- A robot with more than 6 DOF is called **redundant**, it can avoid obstacles. A robot with fewer DOF can reach only some orientations

6. Components - Common Types of Arms and Joints

6.5 The wrist

- Wrist is made of 2 or 3 rotary joints with intersecting axes: **roll, pitch, yaw** (RPY)
- Types: 2 DOF wrist (pitch and roll), 3 DOF wrist (roll-pitch-roll or roll-pitch-yaw)
- The wrist must be light because it sits at the end of the arm and its weight is multiplied by the arm length
- A **spherical wrist** (three axes meet at one point) makes inverse kinematics simple

7. Grippers (End Effectors)

End effector is the device attached to the wrist of the robot to perform the task. It is of two types: **grippers** (hold objects) and **tools** (weld torch, spray gun, drill, grinder).

7.1 Types of grippers by actuation / principle

- **Mechanical grippers** - fingers move to grasp the part (parallel or angular motion)
- **Vacuum (suction) grippers** - suction cups hold flat smooth parts such as glass, sheet metal and boxes
- **Magnetic grippers** - electromagnet or permanent magnet holds ferrous parts
- **Adhesive grippers** - sticky surface for light fabric type parts
- **Hooks, scoops and ladles** - for lifting and for liquids or powders
- **Expandable (inflatable) grippers** - bladder expands in a hole or around a part

7.3 Gripping force

For a friction grip with coefficient of friction mu and number of contact fingers n:

$$F = (m\ g\ a) / (mu\ n)$$

- m = mass of part, g = 9.81, a = acceleration factor (safety factor, usually 2 to 3)
- The force must be large enough to stop the part slipping but small enough not to damage it

7.4 Worked example

Q. A part of mass 5 kg is held by two fingers with friction coefficient 0.25. The part is lifted vertically with an acceleration factor g_factor = 2. Find the gripping force needed per finger.

- Weight W = m g = 5 x 9.81 = 49.05 N
- Using F = W x factor / (mu x n) = 49.05 x 2 / (0.25 x 2) = **196.2 N**

per finger

8. Factors in the Design of Grippers

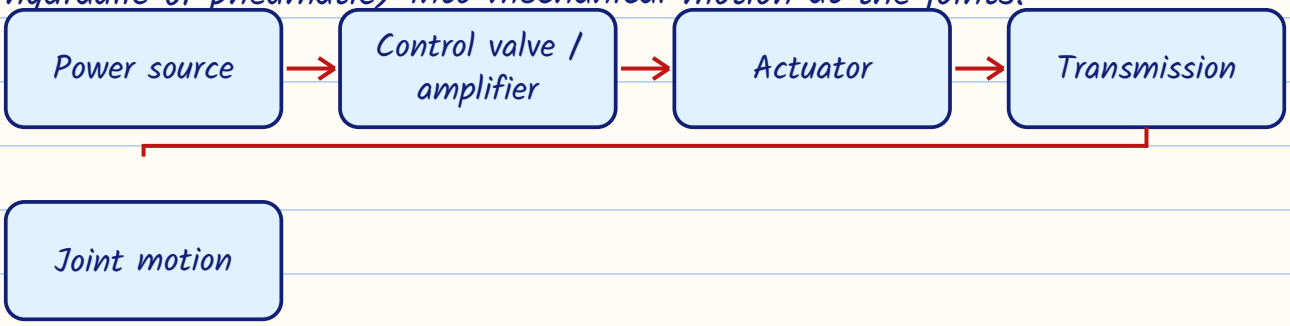
The gripper must hold the part firmly and positively during the entire motion without damaging it, and release it quickly at the right place.

- **Weight and size of the part** - decide the force and the gripper size
- **Shape of the part** - round, flat, prismatic; fingers are shaped to match (V-shaped fingers for cylinders)
- **Gripping force** - enough to overcome weight, inertia and friction with a safety factor
- **Part surface** - hot, wet, oily, soft or fragile parts need special pads
- **Gripper weight** - should be low, the payload of the robot includes the gripper weight
- **Approach and clearance** - fingers must enter and leave without hitting other parts
- **Positive grip** - the part should be located by shape (self centring) so that the position is known
- **Change of part size** - interchangeable fingers or adjustable stroke for part variety
- **Fail safe** - the part must not drop if power fails (spring close, vacuum reservoir)
- **Environment** - temperature, dust, chemicals and explosive atmosphere
- **Cost, reliability and ease of maintenance**
- **Speed of operation** - open and close time must fit within the cycle
- **Use of sensors** - touch or proximity sensors in the fingers to detect part presence and force

9. Robot Actuators and Feedback Components

9.1 Actuators

Actuators are the muscles of the robot. They convert energy (electrical, hydraulic or pneumatic) into mechanical motion at the joints.



- Requirements: high power to weight ratio, high torque at low speed, quick response, smooth motion, accurate control, good efficiency, low cost

9.2 Pneumatic actuators

- Use compressed air at 5 to 8 bar, supplied by a compressor
- Parts: air cylinder (single or double acting), rotary vane motor, directional control valves, FRL unit (filter, regulator, lubricator)
- Mostly used with a bang-bang (end stop) control, so used for pick and place robots and gripper opening
- Advantages: clean, cheap, fast, simple, safe in explosive areas, air is available everywhere
- Limitations: air is compressible so position control is poor, low force, noisy exhaust, needs air preparation

9.3 Hydraulic actuators

- Use mineral oil at 70 to 200 bar from a pump, reservoir and servo valve
- Types: linear cylinders, rotary vane actuators, hydraulic motors (gear, vane, piston)
- Advantages: very high force and torque, high power to weight ratio, stiff, good speed control, self lubricating, can hold position under load
- Limitations: leakage and mess, needs pump, filter and cooler, noise, fire hazard, high maintenance, expensive
- Used in heavy robots such as forging and big painting robots

9.4 Electric actuators

- **DC servo motors** - permanent magnet DC motor, speed controlled by armature voltage, torque is proportional to armature current
- **Brushless DC (BLDC) motors** - electronic commutation, no brush wear, high reliability, widely used now
- **AC servo motors** - induction or synchronous type, driven by variable frequency drives
- **Stepper motors** - move in fixed angular steps for each pulse (see next section)
- Advantages: clean, quiet, accurate, easy to control by computer, good efficiency, low maintenance
- Limitations: lower power to weight than hydraulic, needs gear reducers, sparking in brush types

10. Stepper Motors

A **stepper motor** is a brushless motor that divides a full rotation into a number of equal steps. Each pulse from the driver rotates the shaft by exactly one **step angle**. It runs as an open loop with no feedback sensor.

Step angle = 360 / (number of steps per revolution)

10.1 Types

- **Variable reluctance (VR)** - soft iron toothed rotor, no permanent magnet, step angle 15 degrees or so, no holding torque when off
- **Permanent magnet (PM)** - magnet rotor, bigger steps 45 or 90 degrees, has detent torque
- **Hybrid** - combines both, step angle 1.8 or 0.9 degree, the most common type in robots and CNC

10.3 Speed and position

N (rpm) = (step angle x pulse frequency x 60) / 360

- Position (angle) = number of pulses x step angle
- **Example:** step angle 1.8 degrees, pulse frequency 1000 pulses per second. Steps per revolution = 360 / 1.8 = 200. N = 1000 / 200 x 60 = **300 rpm**
- **Example:** To rotate the shaft by 90 degrees we need 90 / 1.8 = **50 pulses**

10.4 Advantages and limitations

Advantages	Limitations
Open loop control, no sensor needed	Can lose steps if overloaded - error is not known
Accurate positioning and good repeatability	Low torque at high speed, torque falls with speed
Holding torque at standstill	Resonance and vibration at some speeds
Cheap and simple digital control	Low power - used for small robots only
Long life, no brushes	Draws full current even when idle, heating

11. Comparison of Actuators

Feature	Pneumatic	Hydraulic	Electric
Power source	Air 5 to 8 bar	Oil 70 to 200 bar	Electric supply
Power to weight	Low	Very high	Medium
Payload	Small	Very large	Small to medium
Accuracy	Poor	Good	Very good
Speed	High	Medium	High
Stiffness	Low (air compressible)	High	Medium (depends on gearing)
Control	Simple stops	Servo valve	Easy, digital
Cleanliness	Clean	Oil leakage	Very clean
Noise	Exhaust noise	Pump noise	Quiet
Maintenance	Low	High	Low

Cost	Low	High	Medium
Typical use	Pick and place, grippers	Heavy handling, forging	Welding, assembly, general use

Selection rule: light and fast pick and place - pneumatic; very heavy loads - hydraulic; precision and general work - electric servo.

12. Feedback Components - Sensors in Robotics

Sensors give the controller information. **Internal sensors** measure the state of the robot itself (position, velocity, acceleration). **External sensors** measure the environment (touch, proximity, vision, force).

13. Position Sensors

13.1 Potentiometer

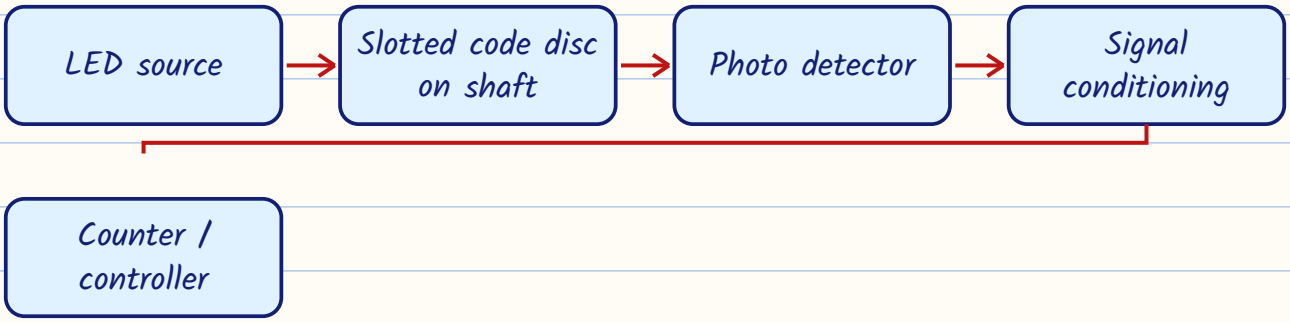
- A resistive track with a sliding contact (wiper) attached to the joint. Output voltage is proportional to the position

$$V_{out} = V_s \times (x / L)$$

- Rotary type for angular and linear type for sliding joints
- Advantages: simple, cheap, analog output directly proportional to position
- Limitations: wear of the wiper and track, noise, limited life, low resolution, friction adds to torque
- **Example:** supply 10 V, a rotary pot with 300 degree travel is at 90 degrees. Vout = 10 x 90 / 300 = **3 V**

13.2 Optical encoders

An encoder is a digital transducer that converts shaft rotation into electrical pulses. It has a **disc with slots**, an **LED light source** and a **photo detector**.



Incremental encoder

- Disc has equally spaced slots. Each slot passes gives one pulse. Counting pulses gives the angle
- Two tracks A and B are shifted by 90 degrees (**quadrature**). The phase tells the direction. A third track Z gives one index pulse per revolution (home mark)
- Resolution = 360 / N degrees per pulse (N = slots). With quadrature counting the resolution is 4 times better
- Limitation: loses the position when power is off - needs a homing routine at start

Absolute encoder

- Disc has multiple concentric tracks coded in **binary** **or** **Gray** **code**. Each position has a unique code
- Resolution = 360 / 2^n for n tracks. Example: 10 tracks = 1024 positions = 0.35 degree
- Advantage: position is known immediately at power on. Limitation: costly and complex
- **Gray** **code** changes only one bit at a time so reading errors are avoided

Example: A 1000 slot incremental encoder with 4x quadrature gives 4000 counts per rev, so the resolution is 360 / 4000 = 0.09 degree.

13.3 Resolver

- A resolver is an electromagnetic rotary transformer, like a small AC machine, with one rotor winding and two stator windings placed 90

degrees apart

- The rotor is fed with an AC reference voltage V sin(wt). The stator outputs are:

$$V_1 = V \sin(wt) \cos(theta),\ V_2 = V \sin(wt) \sin(theta)$$

- The angle is found from the ratio: theta = $\tan^{-1}(V_2 / V_1)$, done by a resolver to digital converter
- Advantages: very rugged, works at high temperature, vibration, dust and oil. Absolute within one turn, long life, no contact wear
- Limitations: needs AC excitation and an RDC circuit, lower resolution than a good encoder, costly

14. Velocity Sensors

- Needed in the servo loop for speed feedback and to damp the motion (the derivative term in PID)

14.1 DC tachogenerator

- A small DC (or AC) generator coupled to the motor shaft. Output voltage is proportional to speed

$$V = K_g \times N$$

- Typical constant 3 to 7 V per 1000 rpm. Simple analog output but brush wear and ripple

15. Tactile Sensors

Tactile (touch) sensors respond to physical contact, force or pressure between the gripper and the object. They give the robot a sense of touch.

15.1 Types

- **Touch sensors** - on/off only (binary). Limit switch, micro switch, whisker that detects contact
- **Force sensors** - measure the size of the force or pressure

- **Slip sensors** - detect the slip of a held object so the grip can be increased

15.2 Principles

- **Strain gauge** - the resistance of a thin wire or foil changes with strain. Bonded to a wrist or finger element and read in a Wheatstone bridge
- **Piezoelectric** - crystals like quartz generate a charge under force. Good for dynamic forces
- **Piezoresistive / conductive rubber** - the resistance of the rubber falls when pressed. Cheap and flexible; used in artificial skin
- **Capacitive** - pressing changes the gap between plates and the capacitance
- **Optical** - pressed rubber changes the light in a fibre or reflection

16. Proximity and Range Sensors

Proximity sensors detect the presence of a nearby object without physical contact. **Range sensors** also measure the distance.

16.1 Inductive proximity sensor

- A coil produces a high frequency magnetic field. A metal object near it causes eddy currents and changes the oscillation, which switches the output
- Detects **metals only**, range 1 to 30 mm, robust against dirt and oil

16.4 Ultrasonic sensor

- Emits a sound pulse of 40 kHz and measures the time of the echo. Distance is found from time of flight

$$d = (v \times t) / 2$$

- Speed of sound v = 343 m/s. **Example:** an echo returns after 5.8 ms. d = 343 x 0.0058 / 2 = **0.995 m, about 1 m**
- Works on any material and colour but has a wide beam and a blind zone near the sensor

17. Quick Revision

- Robot = reprogrammable multifunctional manipulator. 6 DOF are needed for an arbitrary position and orientation
- Configurations: Cartesian PPP, cylindrical RPP, spherical RRP, articulated RRR, SCARA RRP
- Gripper types: mechanical, vacuum, magnetic, adhesive; the design depends on the part weight, shape, force and safety
- Actuators: pneumatic (fast, cheap), hydraulic (powerful), electric (accurate, clean). Stepper = open loop, step angle = 360 / steps
- Position: potentiometer, incremental encoder, absolute encoder, resolver, LVDT. Velocity: tachogenerator, encoder
- Tactile: strain gauge, piezo, conductive rubber. Proximity: inductive, capacitive, optical, ultrasonic

17.1 Important questions

- Define a robot. Explain the main components of an industrial robot with a neat sketch
- Classify robots by configuration. Compare with work envelopes
- What is meant by degrees of freedom? Explain the six DOF of a robot
- Explain the types of joints and the common types of robot arms
- Explain different types of grippers. What factors must be considered while designing a gripper?
- Compare pneumatic, hydraulic and electric actuators
- Explain the working of a stepper motor. Why is it used in open loop control?
- Explain the working of incremental and absolute encoders. Compare them with the resolver
- Explain tactile sensors and proximity sensors used in robots
- A stepper motor has 200 steps per revolution and is driven at 500 pulses per second. Find the speed in rpm (Ans 150 rpm)